CS3000 Algorithms & Data **Practice Quiz 5E**	**Name:** _____ **NUID:** _____

Instructions:

- Please put your phones on silent and in your bag for the duration of the quiz.
- This quiz is designed to last 40 minutes.
- If you have a question, alert a proctor quietly, and we'll be glad to assist.
- If we can't read your answer, we can't grade your answer. Please write legibly.
- Practice version: a mixed review; do it last.

Q1. Short answers. Be precise; a sentence or two each.

(a) What property of the subproblems makes dynamic programming useful, and how does it differ from divide-and-conquer?

(b) The naive recursive `FIB(n)` (base case $n \leq 1$) makes how many calls for $n = 6$? How many *distinct* subproblems are there? How many does memoization actually compute?

(c) Top-down (memoization) and bottom-up (tabulation) compute the same values. Give one reason to prefer bottom-up.

(d) Prove that the 0/1 knapsack recurrence is correct when $w_i \leq w$: show $K(i, w)$ is at least and at most $\max\big(K(i-1, w),\, v_i + K(i-1, w-w_i)\big)$.

(e) The LCS algorithm uses $\Theta(mn)$ space. How can the *length* be computed in $O(n)$ extra space, and why does that break the traceback?

Q1 (continued).

Q2. A warehouse robot packs boxes into a crate with capacity W. There are three box types, and **unlimited copies** of each: type 1: weight 2, value 3; type 2: weight 3, value 5; type 3: weight 5, value 9. Let $U(w)$ be the maximum value that fits in capacity w.

(a) Write a recurrence for $U(w)$ with its base case. Why is a one-dimensional table enough here, when 0/1 knapsack needed $K(i, w)$?

(b) Fill in the table bottom-up for $w = 0, \ldots, 8$.

w	0	1	2	3	4	5	6	7	8
$U(w)$									

(c) For $W = 6$, what is the best value with unlimited copies, and with each type allowed *at most once* (0/1 knapsack)? Explain the difference.

(d) Write bottom-up pseudocode for $U(W)$ and give its running time for n box types.

Q2 (continued).